OKP Research Seminar

# Reliability for multilevel data: A correlation approach

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# Reliability and Measurement Error

- A **reliable** method or measurement instrument is one which gives the same or similar results when it is repeated.

- However, few studies define or report reliability specifically for different hierarchical levels in multilevel data (Geldhof et al., 2014, ten Hove et al., 2022).

- Two recent papers (ten Hove et al., 2022; Schönbrodt et al., 2022) address how to quantify reliability in multilevel settings based on generalizability theory.
  - Nevertheless, both approaches give **inconsistent results** regarding their definition of cluster-level reliability.

# Classical Test Theory

An observed-score $Y_{ij}$ for subject $i$ at occasion $j$ can be decomposed into two parts: a true score $T_{ij}$ and a random measurement error $\epsilon_{ij}$

$$Y_{ij} = T_{ij} + \epsilon_{ij}$$

- where $E[Y_{ij}] = T_{ij}, Cor(\epsilon_{ij}, \epsilon_{ij'}) = Cor(\epsilon_{ij}, T_{ij}) = 0$

The reliability is defined as (Lord et al., 1968)

$$Cor(Y_{ij}, T_{ij})^2 = \frac{\sigma_T^2}{\sigma_Y^2} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$

- It is a ratio of the true-score variance $\sigma_T^2$ over the total variance $\sigma_Y^2$.

# Classical Test Theory

Suppose that there are **two measurements $Y_{ij}$ and $Y_{ij'}$** for the same subject $i$ at the occasions $j$ and $j'$.

- Under the **parallel assumptions**
  - $E[Y_{ij}] = E[Y_{ij'}] = T_i$, $Var(Y_{ij}) = Var(Y_{ij'})$,

- The reliability coefficient is also equal to the correlation between two replicates $Y_{ij}$ and $Y_{ij'}$, i.e.,

$$Cor(Y_{ij}, Y_{ij'}) = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, T_{ij})^2.$$

# Generalizability Theory

## Extending classical test theory

- Recognize and estimate the magnitude of the multiple sources of variation (Cronbach, 1972; Brennan, 2001).
  - such as subjects, raters, items, occasions, etc.
- E.g., a three-way $Subject(S_i) \times Rater(R_j) \times Day(D_m)$ design

| | Rater 1 | | Rater 2 | |
|---|---|---|---|---|
| Subject | Day 1 | Day 7 | Day 1 | Day 7 |
| 1 | $Y_{111}$ | $Y_{112}$ | $Y_{121}$ | $Y_{122}$ |
| 2 | $Y_{211}$ | $Y_{212}$ | $Y_{221}$ | $Y_{222}$ |
| 3 | $Y_{311}$ | $Y_{312}$ | $Y_{321}$ | $Y_{322}$ |
| . | . | . | . | . |
| . | . | . | . | . |
| 10 | $Y_{1011}$ | $Y_{1012}$ | $Y_{1021}$ | $Y_{1022}$ |

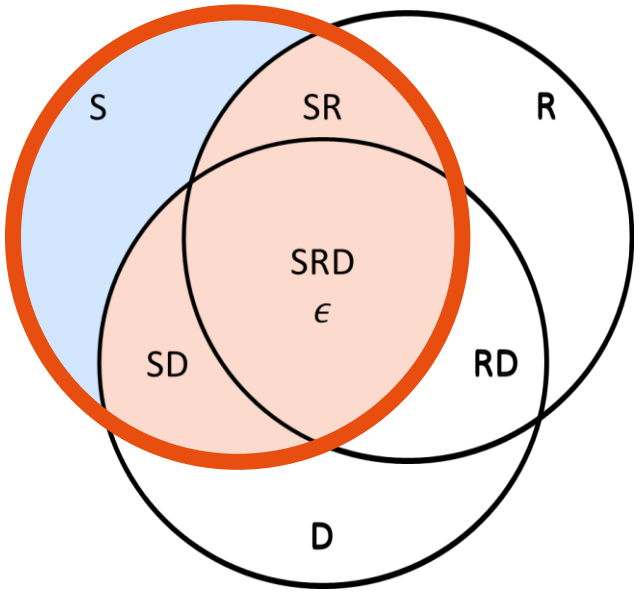
$$Y_{ijt} = \mu + S_i + R_j + D_m + SR_{ij} + SD_{im} + RD_{jm} + \epsilon_{ijm}$$

$$Y_{ijm} = \mu + S_i + R_j + D_m + SR_{ij} + SD_{im} + RD_{jm} + \epsilon_{ijm}$$

## Relative reliability ($R_{rel}$)

- Consistency between subjects' rank-orderings

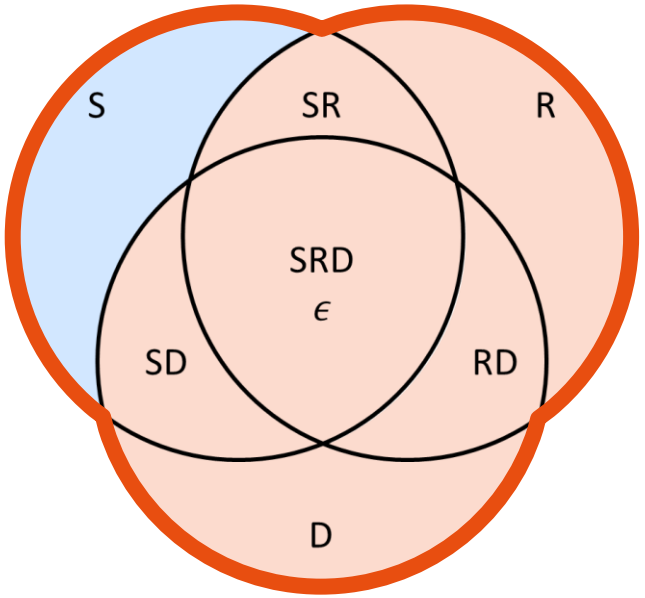
$$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$$




## Absolute reliability ($R_{abs}$)

- Agreement between subjects' absolute scores

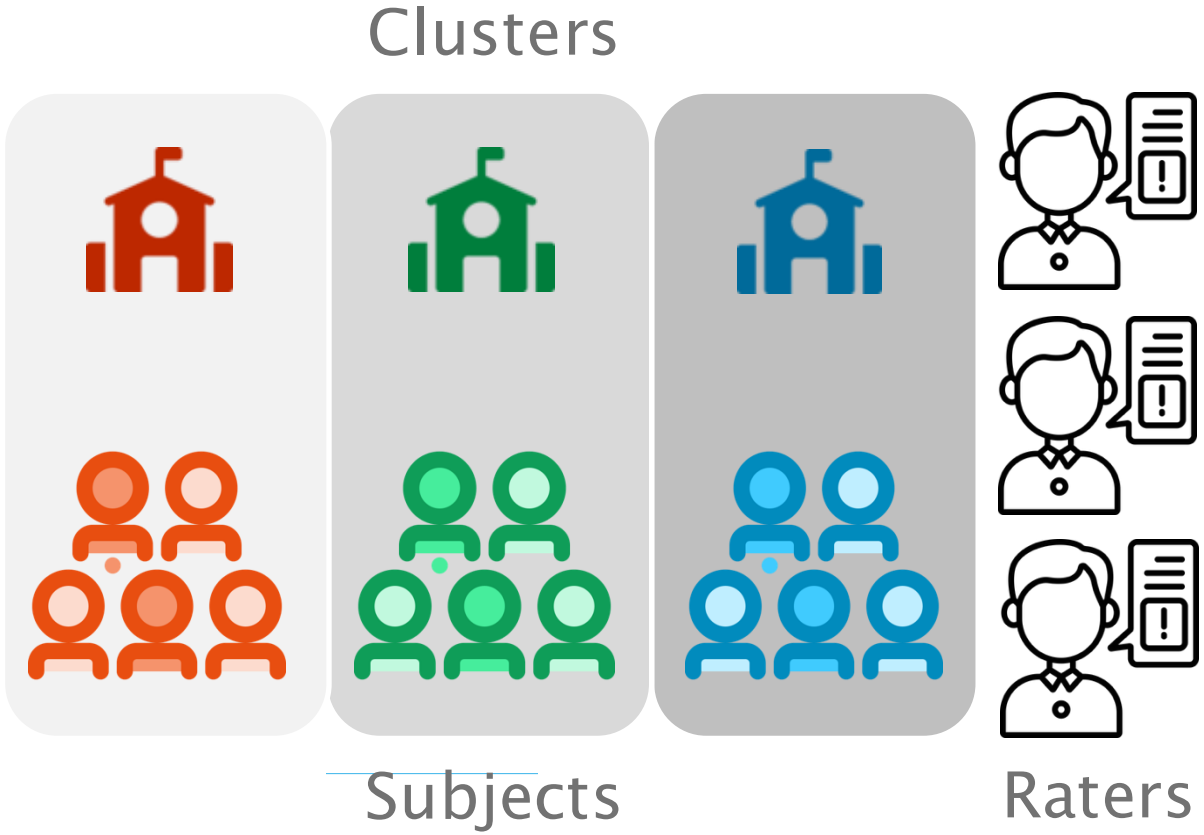
$$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$$



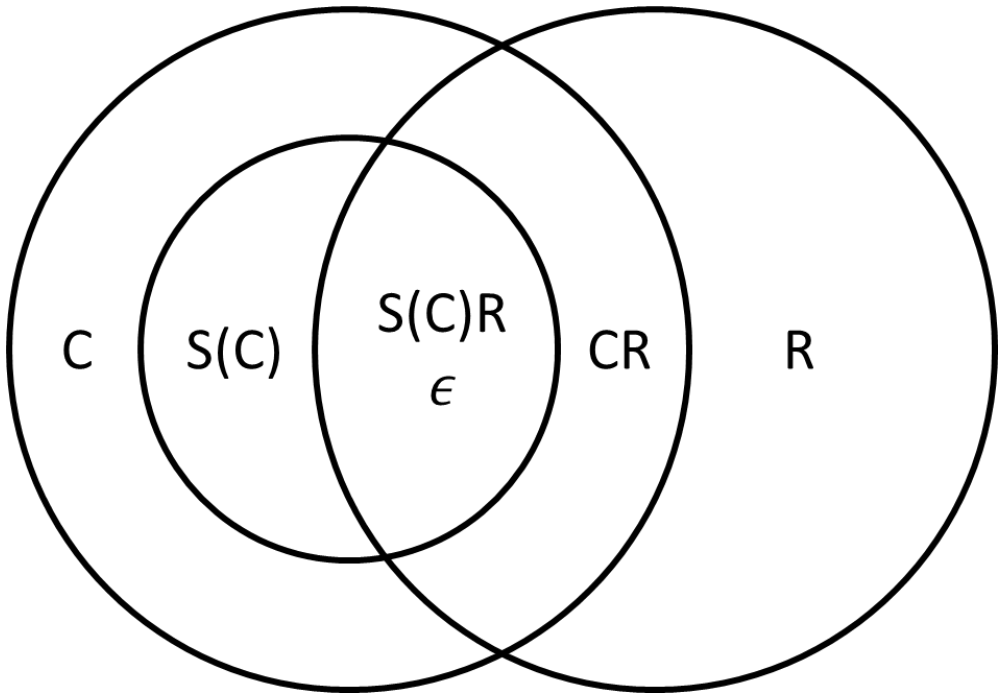
# Nested Structures

## Case 1: $Subject(Cluster) \times Rater$ design

- subjects ($S_{i(k)}$) are nested within clusters ($C_k$), and
- raters ($R_j$) are crossed with $S$ & $C$.



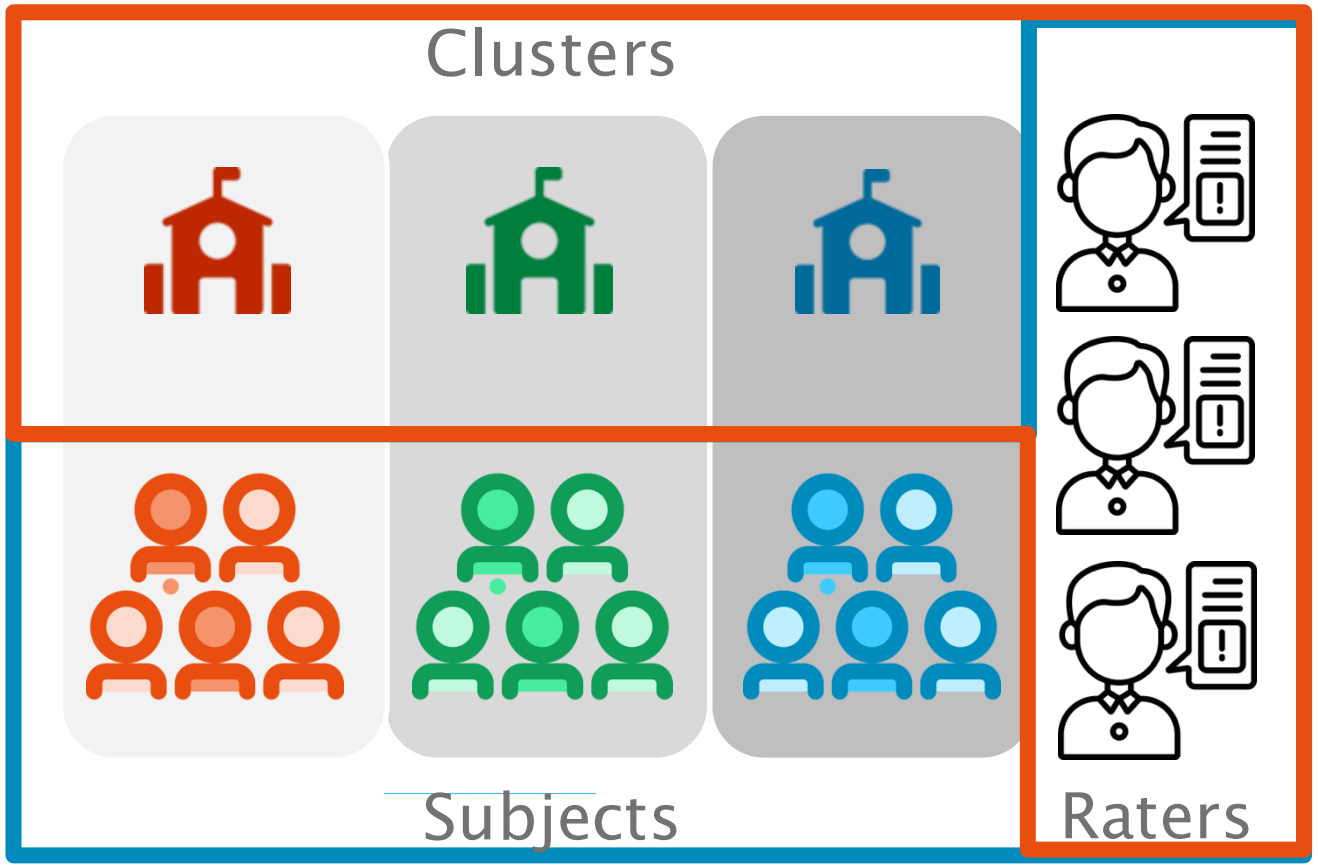

$$Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$$



# Nested Structures

**Case 1:** $Subject(Cluster) \times Rater$ design

- subjects ($S_{i(k)}$) are nested within clusters ($C_k$), and
- raters ($R_j$) are crossed with $S$ & $C$.




Reliability for
the different levels?
Subject / Cluster

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# Interrater Reliability for Multilevel Data: A Generalizability Theory Approach

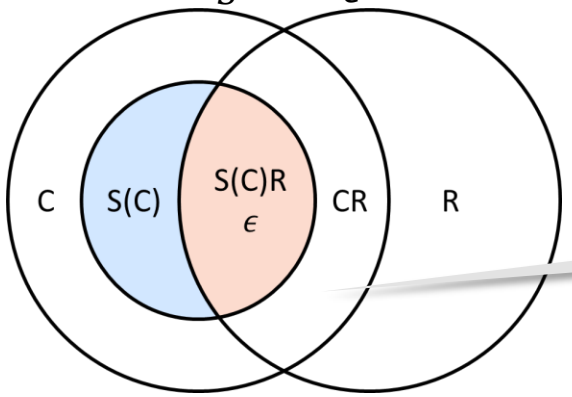
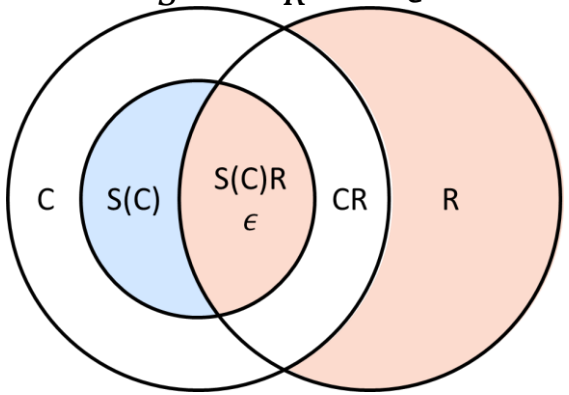
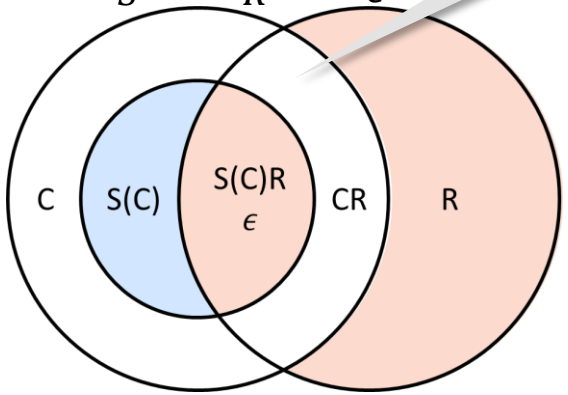
Debby ten Hove, Terrence D. Jorgensen, and L. Andries van der Ark
Research Institute of Child Development and Education, University of Amsterdam

# Measuring motivational relationship processes in experience sampling: A reliability model for moments, days, and persons nested in couples

Behavior Research Methods (2022) 54:1869–1888
https://doi.org/10.3758/s13428-021-01701-7

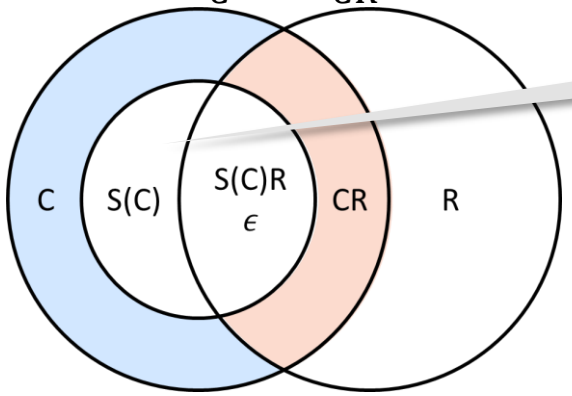
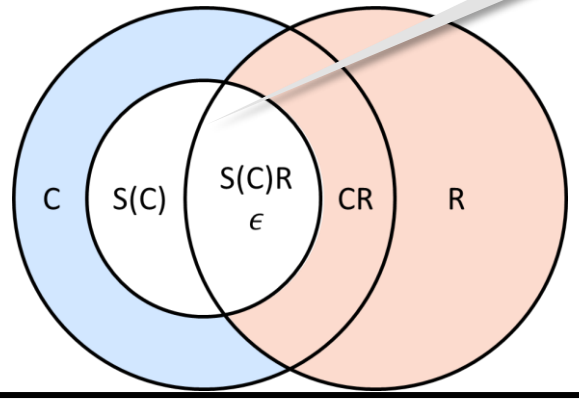
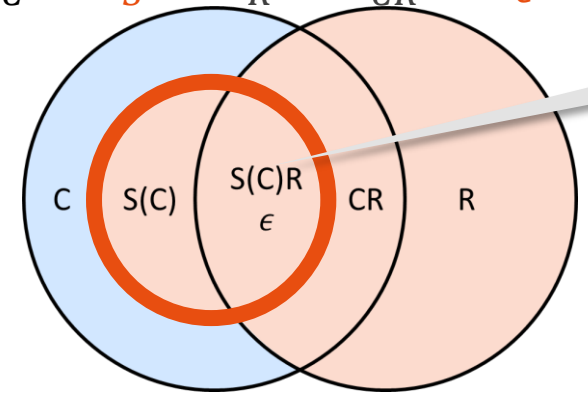
Felix D. Schönbrodt[1] · Caroline Zygar-Hoffmann[1] · Steffen Nestler[2] · Sebastian Pusch[3] · Birk Hagemeyer[3]

# Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) |
|---|---|---|
| **Subject-level** | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ | (not defined) |
| $R^S_{rel}$ |   | |
| $R^S_{abs}$ | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$   | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$   |

Ignore cluster-level components

# Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) |
|---|---|---|
| **Cluster-level** | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$ | (not defined) |
| | $R^C_{rel}$   | |
| | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$ | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$ |
| | $R^C_{abs}$   |   |

Ignore subject-level components

Include everything

# Comparison

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) |
|---|---|---|
| Subject level | Do not contain any cluster-related (higher level) variance components (i.e., $\sigma_C^2$, $\sigma_{CR}^2$) | |
| Cluster level | Do not contain any subject-related variance (lower level) components (i.e., $\sigma_{S(C)}^2, \sigma_{S(C)R}^2$) | Contain subject-related variance components |
| Types | Separate **relative** and **absolute** reliability | Only discuss **absolute** reliability |

# Correlation Approach for Multilevel Data

The reliability coefficient can be defined as the **correlation** between two measurements from the same subject $i$ on different raters (or occasions, etc.) $j$ and $j'$ (Vangeneugden et al., 2005).

| | Subject-level | Cluster-level |
|---|---|---|
| Relative reliability | $\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j')$ | $\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j')$ |
| Absolute reliability | $\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k)$ | $\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k))$ |

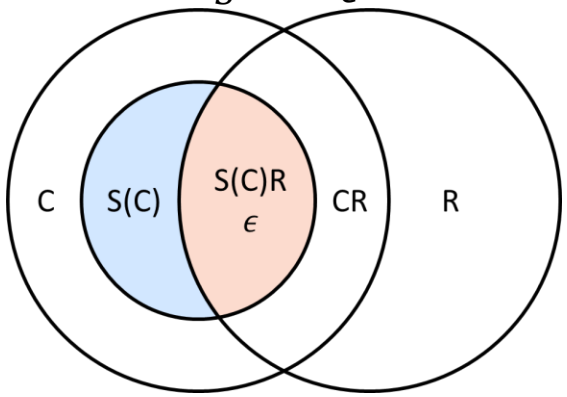
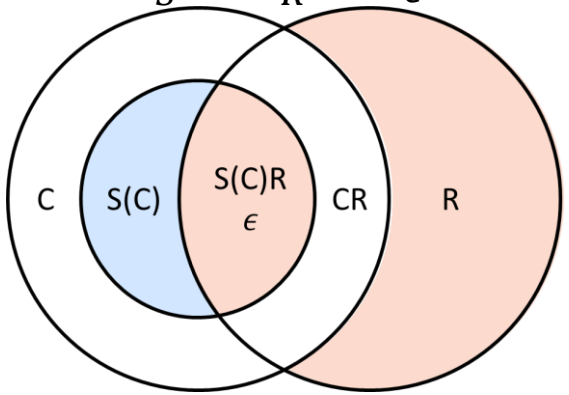
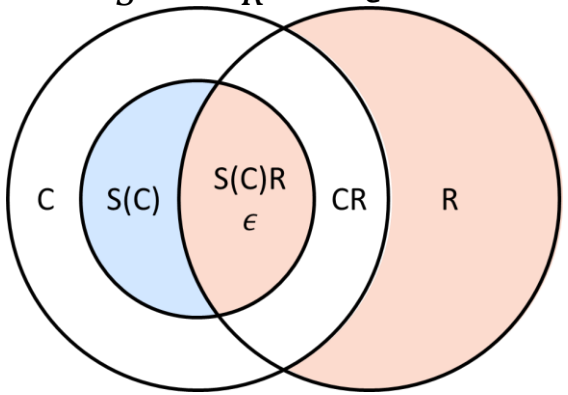
$C_k, R_j, R_{j'}, CR_{kj}, CR_{kj'}$

# Correlation Approach for Multilevel Data

The reliability coefficient can be defined as the **correlation** between two measurements from the same subject $i$ on different raters (or occasions, etc.) $j$ and $j'$ (Vangeneugden et al., 2005).

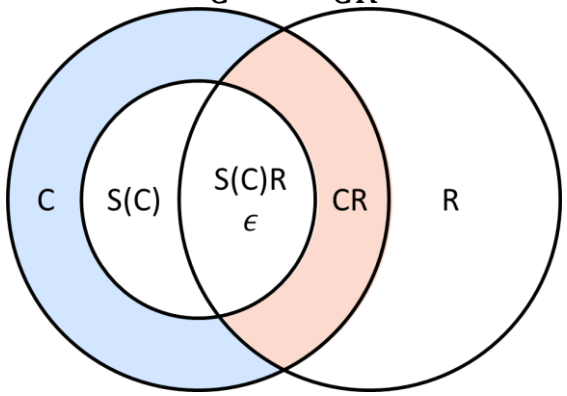
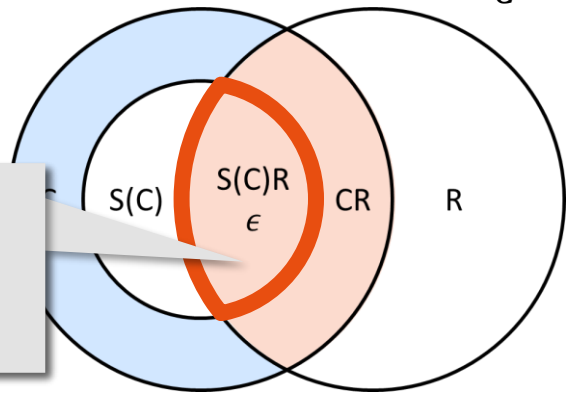
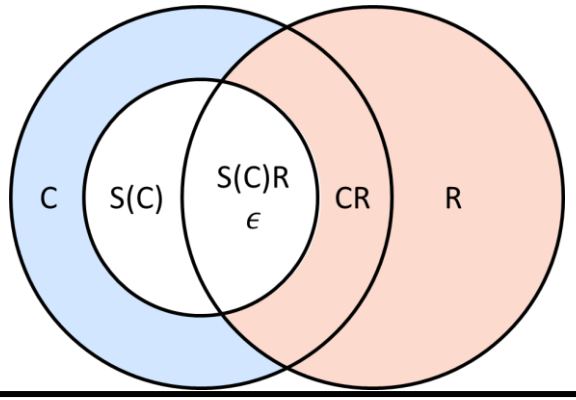
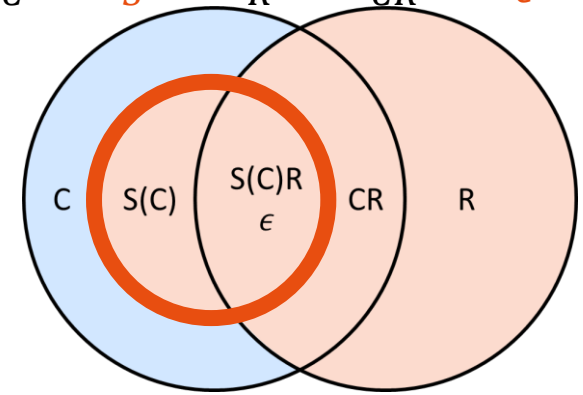
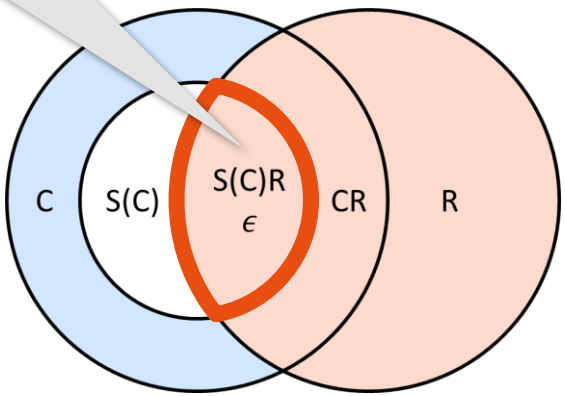
| | Subject-level | Cluster-level |
|---|---|---|
| Relative reliability | $\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j')$ | $\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j')$ |
| Absolute reliability | $\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k)$ | $\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k))$ |

# Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) |
|---|---|---|
| **Subject-level** | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ | (not defined) |
| $R^S_{rel}$ |   | |
| $R^S_{abs}$ | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$   | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$   |

Because $CR$ is random and influences rater effects

# Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

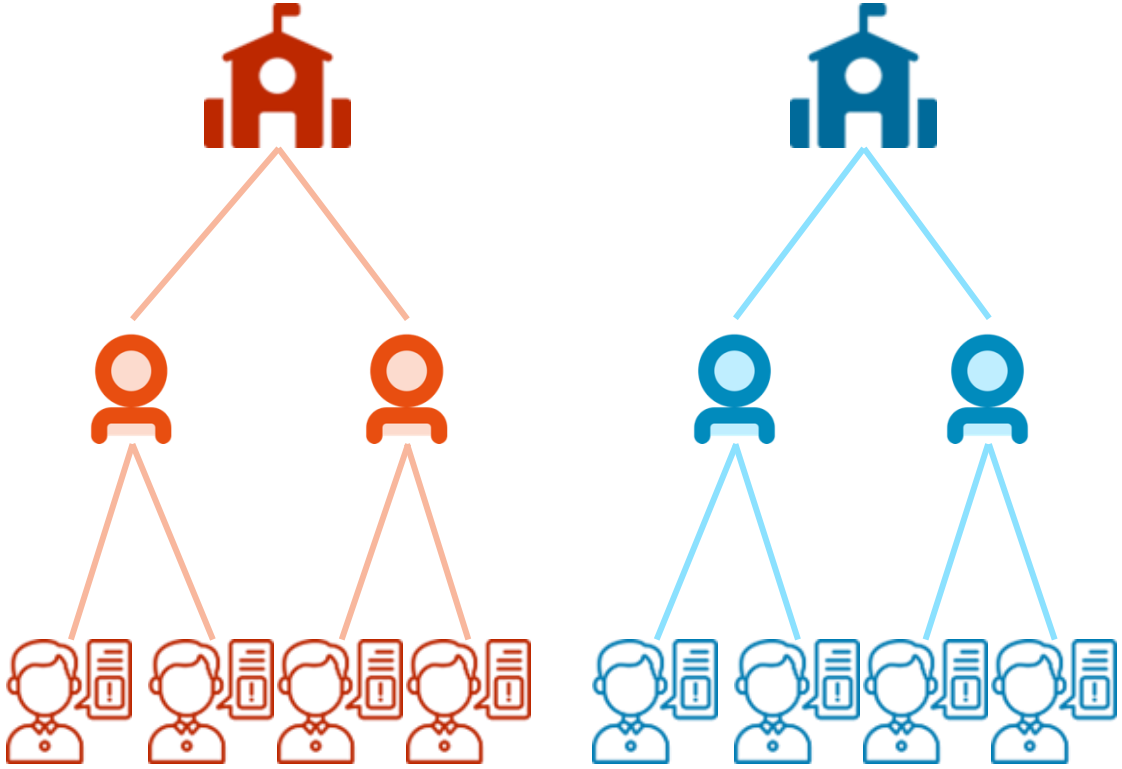
| | ten Hove et al. (2022) | Schönbrodt et al. (2022) | Correlation approach |
|---|---|---|---|
| **Cluster-level** | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$ | (not defined) | $\mathrm{Cor}\left( Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j' \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$ |
| $R^C_{rel}$ |  | |  |
| $R^C_{abs}$ | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$  | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$  | $\mathrm{Cor}\left( Y_{i(k)j}, Y_{i(k)j'} \mid i(k) \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$  |

With measurement error variance

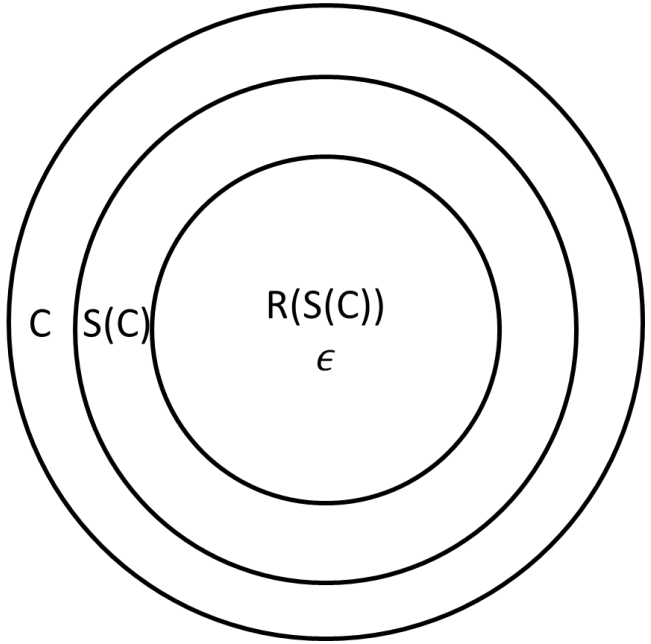
# Another Nested Structure Example

**Case 2:** *Rater(Subject(Cluster))* design

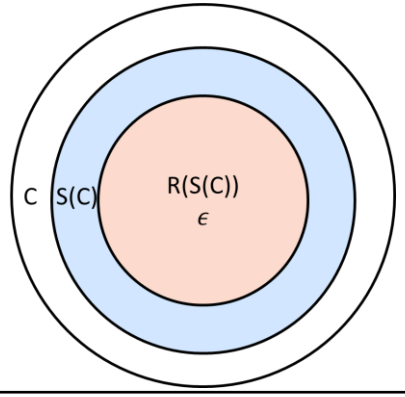
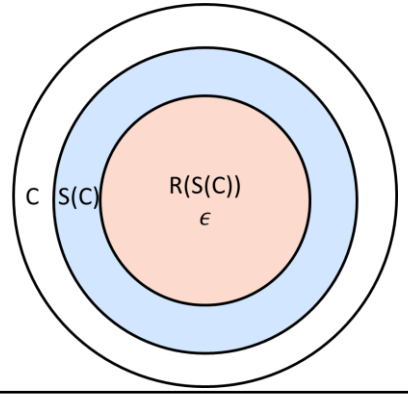
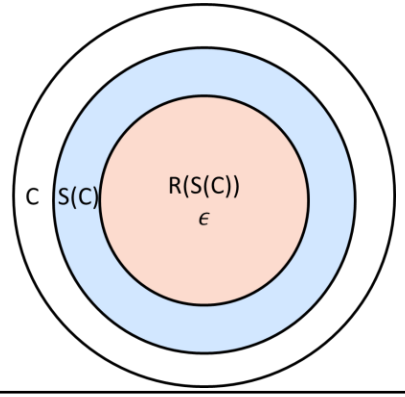
- subject ($S_{i(k)}$) are nested within clusters ($C_k$), and
- raters ($R_{j(i(k))}$) are also nested within S and C



$$Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$$



# Case 2: $Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$

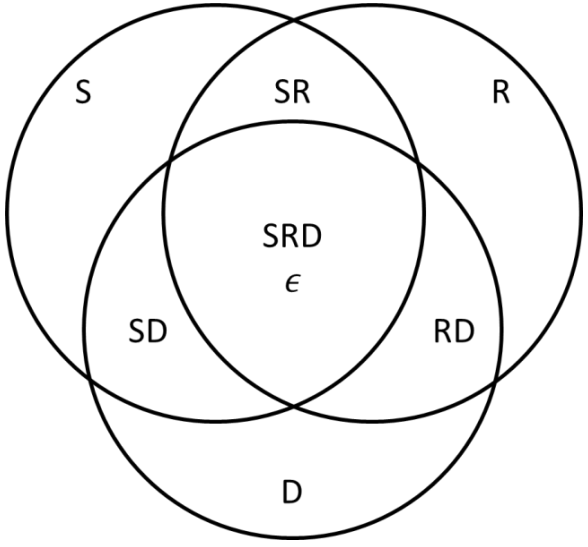
| | ten Hove et al. (2022) | Schönbrodt et al. (2022) | Correlation approach |
|---|---|---|---|
| **Subject-level** | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ | $\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ |
| $R^S$ |   |   |   |
| **Cluster-level** | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_\epsilon^2}$ | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_\epsilon^2}$ | $\mathrm{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$ |
| $R^C$ | Cannot be defined |   |   |

# Advantages of the Correlation Approach

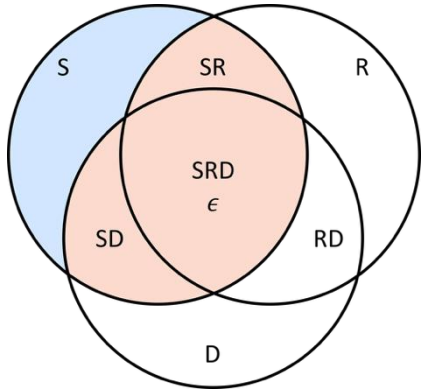
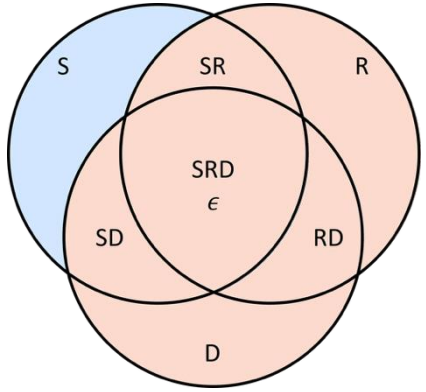
- Various types of reliability coefficients can be defined by similar correlation formulas.
  - Relative vs. Absolute
  - Subject-level vs. Cluster-level
  - Overall vs. Test-retest vs. Interrater

- Recall the three-way $Subject(S_i)\times Rater(R_j)\times Day(D_t)$ design

$$Y_{ijt} = \mu + S_i + R_j + D_t + SR_{ij} + SD_{it} + RD_{jt} + \epsilon_{ijt}$$

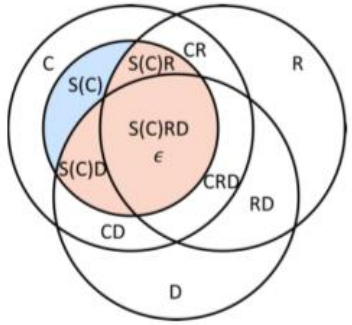
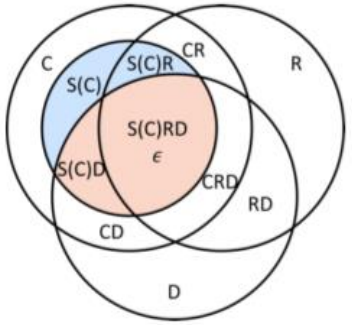
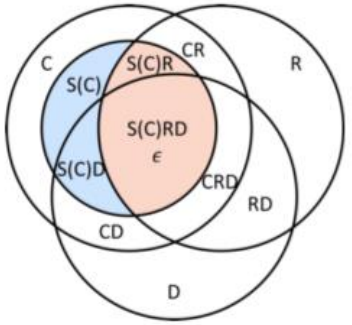
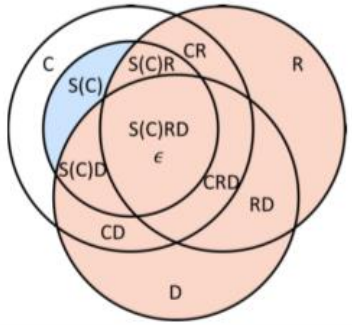
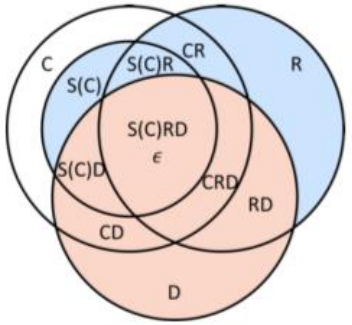
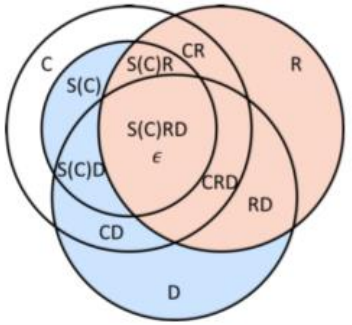


# $Subject(S_i)\times Rater(R_j)\times Day(D_t)$ design

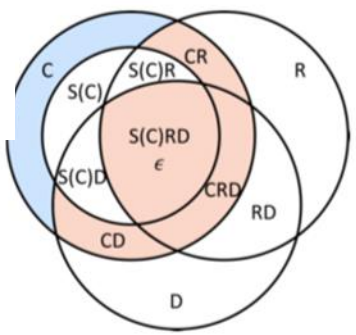
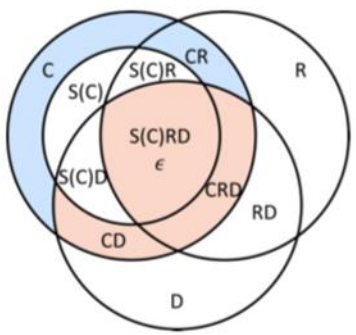
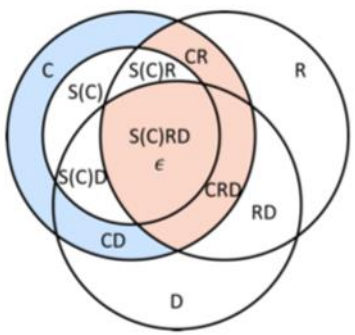
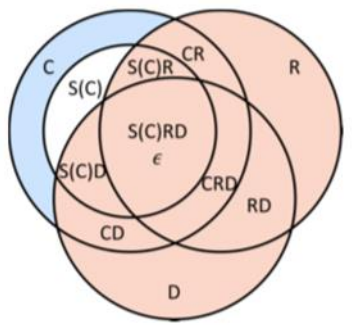
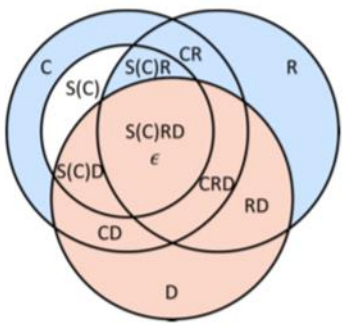
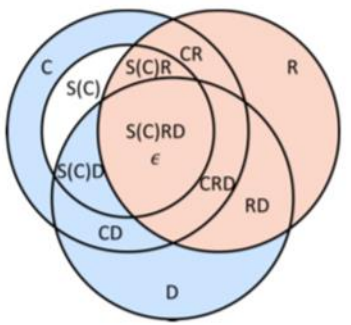
| Overall | Test-retest | Interrater |
|---|---|---|
| $\text{Cor}\left(Y_{ijm}, Y_{ij'm'} \mid j, j', m, m'\right)$ | $\text{Cor}\left(Y_{ijm}, Y_{ijm'} \mid j, m, m'\right)$ | $\text{Cor}\left(Y_{ijm}, Y_{ij'm} \mid j, j', m\right)$ |
| $R_{rel}$  |  |  |
| $\text{Cor}(Y_{ijm}, Y_{ij'm'})$ | $\text{Cor}(Y_{ijm}, Y_{ijm'})$ | $\text{Cor}(Y_{ijm}, Y_{ij'm})$ |
| $R_{abs}$  |  |  |

# $Subject(Cluster)\times Rater\times Day$ design

## Subject-level

| Reliability | Overall | Test-retest | Interrater |
|---|---|---|---|
| | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm'} \mid k, j, j', m, m')$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)jm'} \mid k, j, m, m')$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm} \mid k, j, j', m)$ |
| $R^{S}_{rel}$ |  |  |  |
| | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm'} \mid k)$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)jm'} \mid k)$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm} \mid k)$ |
| $R^{S}_{abs}$ |  |  |  |

## Cluster-level

| Reliability | Overall | Test-retest | Interrater |
|---|---|---|---|
| | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm'} \mid i(k), j, j', m, m')$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)jm'} \mid i(k), j, m, m')$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm} \mid i(k), j, j', m)$ |
| $R^{C}_{rel}$ |  |  |  |
| | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm'} \mid i(k))$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)jm'} \mid i(k))$ | $\mathrm{Cor}(Y_{i(k)jm}, Y_{i(k)j'm} \mid i(k))$ |
| $R^{C}_{abs}$ |  |  |  |

# Advantages of the Correlation Approach

- Various types of reliability coefficients can be defined by similar correlation formulas.
  - Relative vs. Absolute
  - Subject-level vs. Cluster-level
  - Overall vs. Test-retest vs. Interrater

- The correlation is easy to interpret.
  - It reflects the similarity of scores for the same person under various conditions

- The reliability coefficient obtained by calculating the correlation under ANOVA models can be expressed as a ratio of variances
  - Indicate which sources of variance contribute to the **systematic variation** and which ones serve as the **unsystematic noise**.

# Discussions

- Point out the different reliability definitions for multilevel data in the latest literature (ten Hove et al., 2022; Schönbrodt et al., 2022).
  - there is no consensus on defining the reliability coefficient

- Using correlation is a promising approach to define reliability for multilevel settings.
  - a principal rule to derive our formula
  - easy to understand and has the flexibility for complicated models, e.g., (generalized) linear mixed-effects models (Molenberghs et al. 2007)

# Discussions

- Challenges and limitations
  - reliability coefficients for linear mixed-effects models are inconsistencies between ICC, GT, and the correlation approach
  - the meaning of the conditioning
  - the ANOVA model has too many assumptions

- Future work
  - consider the role of fixed effects when we discuss reliability
  - explore the statistical properties of and the estimation method for reliability defined by the correlation approach
  - need an empirical study

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Q&A

# Reliability for multilevel data: A correlation approach

**Tzu-Yao Lin**, Francis Tuerlinckx, Sophie Vanbelle

Reliability for other nested structure lists in the appendix slides

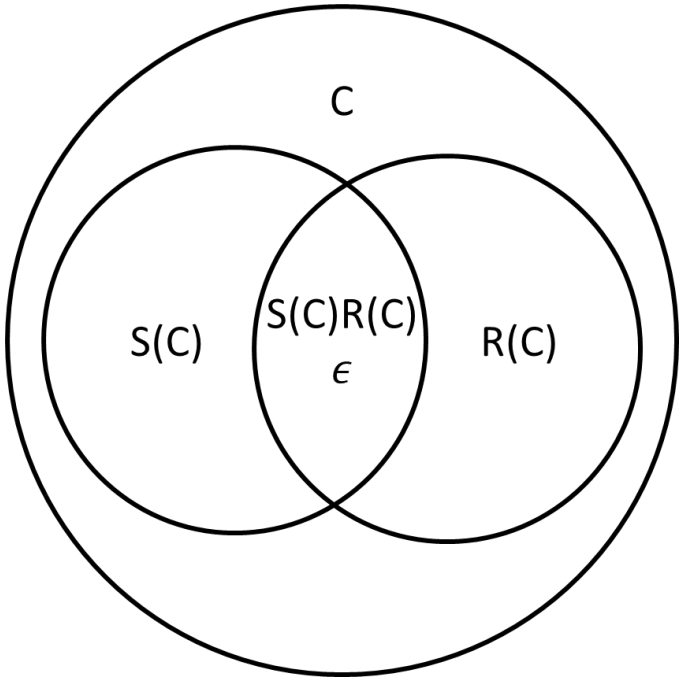
Maastricht University KU LEUVEN

12-03-2024

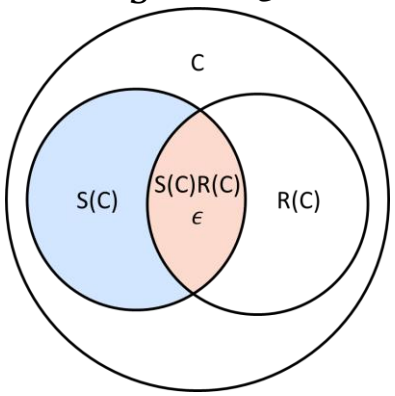
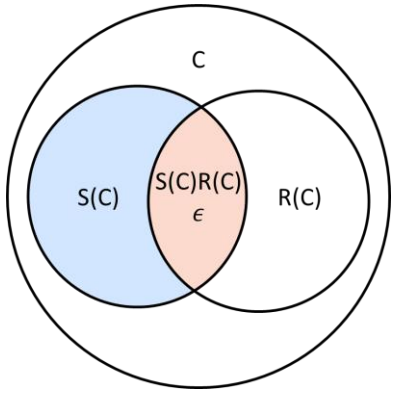
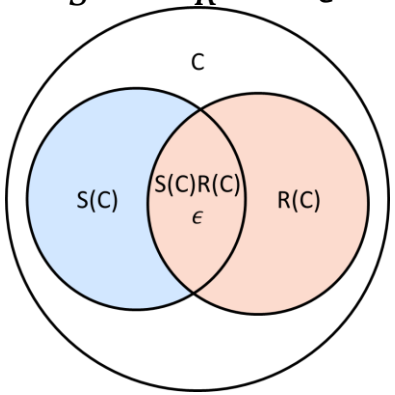
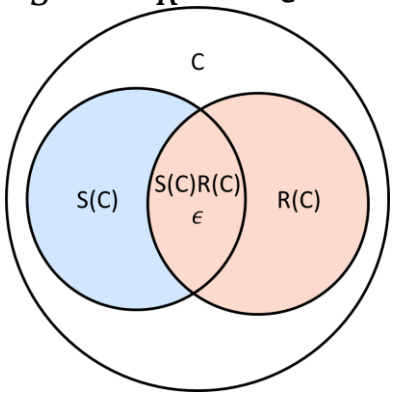
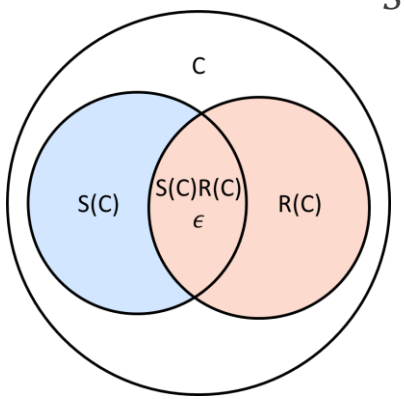
# Other Nested Structure Examples

- Case 3: Subjects (S) are nested within clusters (C),
  Raters (R) are *nested within* C and *crossed with* S

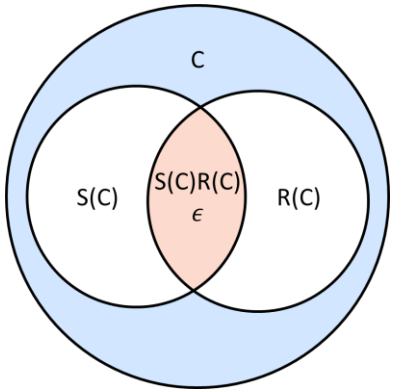
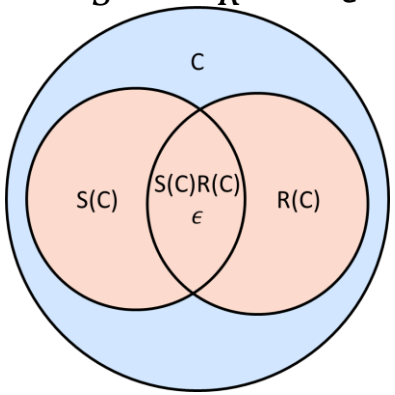
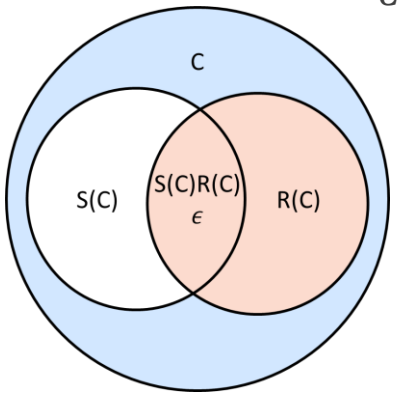
$$Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$$



# Case 3: $Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) | Correlation approach |
|---|---|---|---|
| **Subject-level** | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ | | $\mathrm{Cor}\left( Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid k, j, j' \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ |
| $R^S_{rel}$ |  | (not defined) |  |
| | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ | $\mathrm{Cor}\left( Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ |
| $R^S_{abs}$ |  |  |  |

# Case 3: $Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) | Correlation approach |
|---|---|---|---|
| **Cluster-level** | | | $\mathrm{Cor}(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid i, j, j') = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$ |
| $\mathrm{R}^C_{rel}$ | Cannot be defined because it ignores all lower-level variances | (not defined) |   |
| $\mathrm{R}^C_{abs}$ | | $\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$   | $\mathrm{Cor}(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid i) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_\epsilon^2}$   |

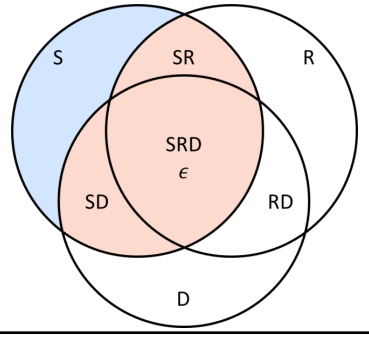
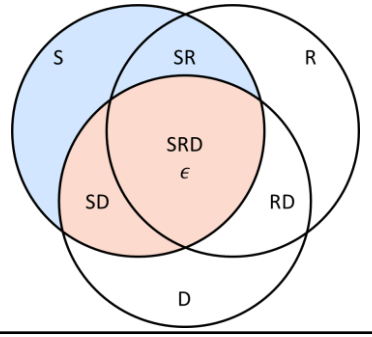
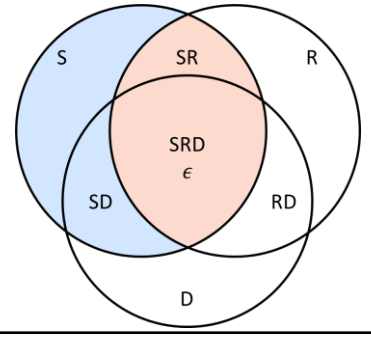
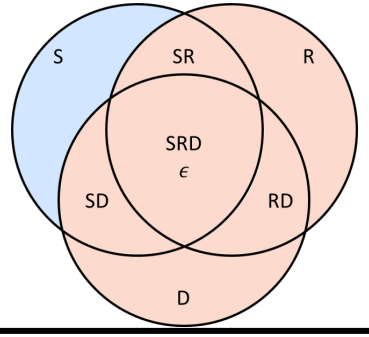
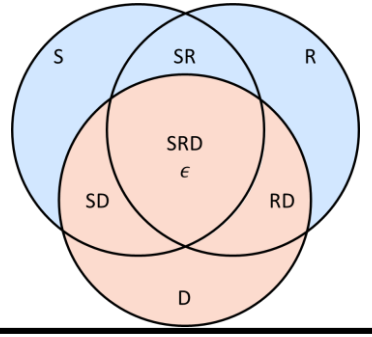
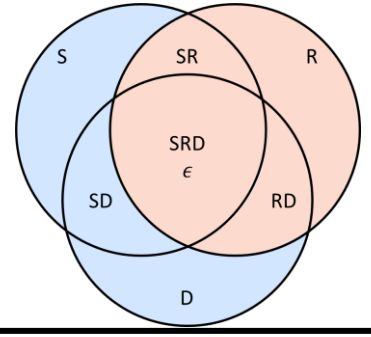
# Intraclass Correlation Coefficient (ICC)

ICC is a widely used reliability coefficient in interrater reliability analyses (Bartko 1966; McGraw & Wang 1996; Shrout & Fleiss, 1979).

- Different forms of ICCs

| | ICC | One-way $Y_{j(i)} = \mu + S_i + \epsilon_{j(i)}$ | Two-way random effect $Y_{ij} = \mu + S_i + R_j + \epsilon_{ij}$ | Two-way mixed effect $Y_{ij} = \mu + S_i + r_j + \epsilon_{ij}$ |
|---|---|---|---|---|
| Single | Relative Consistency | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2} = Cor(Y_{j(i)}, Y_{j'(i)})$ | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, Y_{ij'}\|j, j')$ | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$ |
| | Absolute Agreement | | $\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, Y_{ij'})$ | $\frac{\sigma_S^2}{\sigma_S^2 + \theta_R^2 + \sigma_\epsilon^2}$ |

# GT vs. Correlation Approach

| Reliability | Overall | Test-retest | Interrater |
|---|---|---|---|
| Relative Consistency | $\mathbf{GC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'} \mid j, j', t\ t') = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$   | $\mathrm{Cor}(Y_{ijt}, Y_{ijt'} \mid j, t\ t') = \frac{\sigma_S^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$   | $\mathrm{Cor}(Y_{ijt}, Y_{ij't} \mid j, j', t) = \frac{\sigma_S^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$   |
| Absolute Agreement | $\mathbf{DC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'}) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$   | $\mathrm{Cor}(Y_{ijt}, Y_{ijt'}) = \frac{\sigma_S^2 + \sigma_R^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$   | $\mathrm{Cor}(Y_{ijt}, Y_{ij't}) = \frac{\sigma_S^2 + \sigma_D^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$   |

# Comparison

| | ten Hove et al. (2022) | Schönbrodt et al. (2022) |
|---|---|---|
| Variance decomposition | Generalizability theory | |
| Reliability coefficient | In terms of ICCs | Extend from Shrout & Land's (2012) formulas which are based on the DC |
| Subject level | Do not contain any cluster-related (higher level) variance components (i.e., $\sigma_C^2$, $\sigma_{CR}^2$) | |
| Cluster level | Do not contain any subject-related variance (lower level) components (i.e., $\sigma_{S(C)}^2, \sigma_{S(C)R}^2$) | Contain subject-related variance components |
| Types | Separate *relative* and *absolute* reliability | Only discuss *absolute* reliability |
| Other characteristics | • Discuss subject-level and cluster-level reliability in three kinds of nested structure | • Discuss the roles of the *fixed effects*<br>• Also build up the within-subject reliability under two different time scales |